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1 using LinearAlgebra
```

Section 2.3 Problem 4

If n = 100: The forward error is **1.9588684008198243e-9**. The backward error is **2.7284841053187847e-12**. The relative forward error is **1.9588684008198243e-9**. The relative backward error is **5.484527758910262e-16**. the error magnification factor is **3.571626376833286e6**. Finally the condition number is **6.18430776032762e7**.

If n = 200: The forward error is **8.763575214398855e-7**. The backward error is **1.4551915228366852e-11**. The relative forward error is **8.763575214398855e-7**. The relative backward error is **7.293377484321715e-16**. the error magnification factor is **1.2015798213156478e9**. Finally the condition number is **1.2906621977890022e10**.

If n = 300: The forward error is **3.530838662957336e-5**. The backward error is **3.2741809263825417e-11**. The relative forward error is **3.530838662957336e-5**. The relative backward error is **7.28717297439084e-16**. the error magnification factor is **4.8452790613941635e10**. Finally the condition number is **6.203930120211332e11**.

If n = 400: The forward error is **0.0013277521890622257**. The backward error is **1.1641532182693481e-10**. The relative forward error is **0.0013277521890622257**. The relative backward error is **1.4568300073976043e-15**. the error magnification factor is **9.113981606090365e11**. Finally the condition number is **1.4752813090246375e13**.

If n = 500: The forward error is **0.020800043702734028**. The backward error is **1.1641532182693481e-10**. The relative forward error is **0.020800043702734028**. The relative backward error is **9.32143695675338e-16**. the error magnification factor is **2.2314203056068945e13**. Finally the condition number is **2.289719699894379e14**.

```
1 begin
2
3     result = ""
4
5     for i in 1:5
6         n = 100 * i
7         M = [sqrt((i - j)^2 + n/10) for i in 1:n, j in 1:n]
8         Minv = inv(M)
9         x = ones(n, 1)
10        b = M * x
11
12        x_c = M\b
13
14        FE = norm(x - x_c, Inf)
15        rFE = FE/norm(x, Inf)
16        BE = norm(b - M*x_c, Inf)
17        rBE = BE/norm(b, Inf)
18
19        global result *= ""**If n = $(n)**:
20
```

The forward error is FE . The backward error is BE . The relative forward error is rFE . The relative backward error is rBE . the error magnification factor is rFE/rBE . Finally the condition number is $opnorm(M, Inf) * opnorm(Minv, Inf)$.

"""

end

Markdown.parse(result)

end